Node Class Documentation

Source File: ‘`Node.h`’
Namespace: dsp
Class Header: `template <class T> class Node`

Overview

The *Node* class is a generic doubly linked node.

Constructors

- `Node()` (default constructor)
  - **Purpose:** Assigns null to its links and the default value to its data.
- `Node(const T& obj)`
  - **Purpose:** Assigns *obj* to its data and null to its links.
  - **Parameter(s):**
    - *lhs*: Constant *T* reference object.
- `Node(const Node<T>& obj)` (copy constructor)
  - **Purpose:** Copies the data of *obj* and assigns null to its links.
  - **Parameter(s):**
    - *obj*: Constant *Node* reference object.

Assignment Operators

- `operator=(const Node<T>& rhs)`
  - **Purpose:** Copies the data of *rhs* only.
  - **Parameter(s):**
    - *rhs*: Constant *Node* reference object.
  - **Return:** `*this`.

Destructor

- `~Node()`
  - **Purpose:** Does nothing.

Methods

- `data() const`
  `data()`
  - **Purpose:** Provides access to the data field.
  - **Return:** *T* reference object.
- `prev() const`
  `prev()`
  - **Purpose:** Provides access to the previous link field.
  - **Return:** *Node* pointer reference object.
- `next() const`
  `next()`
  - **Purpose:** Provides access to the next link field.
  - **Return:** *Node* pointer reference object.

Non-Member Functions

- `clone(Node<T>* obj)`
  - **Purpose:** Returns a copy of *obj*.
  - **Parameter(s):**
    - *obj*: *Node* pointer object.
  - **Return:** *Node* pointer object.
- `clear(Node<T>*& obj)`
  - **Purpose:** Deletes the linked list pointed to by *obj*.
  - **Parameter(s):**
    - *obj*: *Node* pointer reference object.